b. Disadvantages

i. Non-linear characteristics.
ii. Low-voltage only generated.
iii. Least – sensitivity to temperature changes.

7.8 Photo voltaic Cells (Solar cells)

Photo voltaic cells may be used in a number of applications. The silicon solar cell converts the radiant energy of the sun into electrical power. The solar cell consists of a thin slice of single crystal p-type silicon, up to 2cm square, into which a very thin layer of n-type material is diffused. The conversion efficiency depends on the spectral content and intensity of the illumination.

a. Advantages

i. Simple and green environment.
ii. Pollution free.
iii. Renewable electric power generation.



Fig. 7.15 **Photo voltaic Cell**

iv. Use batteries to store extra power for use at night.
v. Solar can be used to heat water, power homes and building, even power cars.
vi. Safer than traditional electric current.

b. Disadvantages

i. High initial costs for material and installation.
ii. Needs a lot of space.
iii. No solar power at night. Hence there is a need for a large battery bank.
iv. Cloudy days do not produce much energy.

Glossary

Piezo resistivity	-	தகவுத்தடை
Potentiometer	-	மின்னழுத்தமானி
Signal	-	சிக்னல்
Strain gauge	-	திரிபளவுமானி
Thermistor	-	வெப்பத்தடையகம்
Permeability	-	ஊடுருவும் தன்மை
Linear variable differential transformer	-	நேரியல் மாறி மாறுபட்டமின்மாற்றி
Thermocouple	-	வெப்பமின்னிரட்டை

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